



2024-25

GRAVITATION

Gravitational Force

There exists a force of attraction between any two particles of matter in the universe such that the force depends only on the masses of the particles and the separation between them. It is called the gravitational force and the mutual attraction is called gravitation.

The gravitational force between the earth and the moon. The gravitational force is a universal force, i.e., it acts between any two objects in the universe.

Centripetal Force

In uniform circular motion of a body, the force acting on the body is directed towards the centre of the circle. This force is called centripetal force.

Thus, a force acts on any object moving along a circle and it is directed towards the centre of the circle. This is called the centripetal force. 'Centripetal' means centre seeking, i.e., the object tries to go towards the centre of the circle because of this force.

Example:

The moon revolves around the earth due to the gravitational force exerted on it by the earth. This force is directed towards the centre of the earth and is thus a centripetal force.

Newton's Law of Motion

First law:

An object continues to remain at rest or in a state of uniform motion along a straight line unless an external unbalanced force acts on it.

Second law:

The rate of change of momentum is proportional to the applied force and the change of momentum occurs in the direction of the force.

Third law:

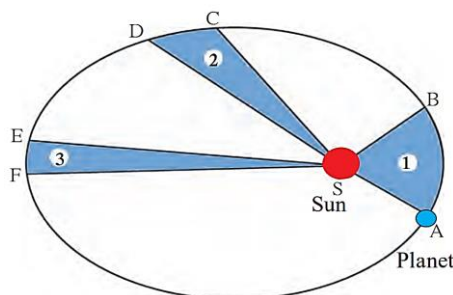
Every action force has an equal and opposite reaction force which acts simultaneously.

[Equal in magnitude and opposite in direction.]

Kepler's Laws of Planetary Motion

First law:

The orbit of a planet is an ellipse with the Sun at one of the foci.



The orbit of a planet moving around the Sun.

Figure shows the elliptical orbit of a planet revolving around the Sun. S denotes the position of the Sun.



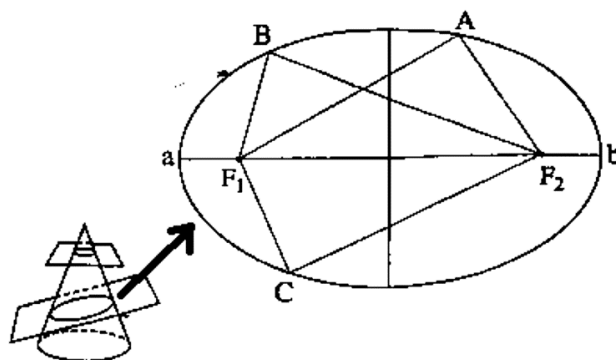
Second law:

The line joining the planet and the Sun sweeps equal areas in equal intervals of time.

Third law:

The square of the period of revolution of a planet around the Sun is directly proportional to the cube of the mean distance of the planet from the Sun.

$$(\text{Period of revolution})^2 \propto \left(\frac{ab}{2}\right)^3$$



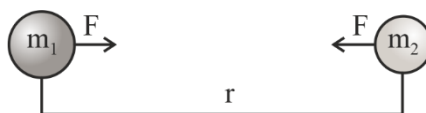
Thus, if r is the average distance of the planet from the Sun and T is its period of revolution, then,

$$T^2 \propto r^3,$$

$$\text{i.e., } T^2/r^3 = \text{constant} = K.$$

Mathematical form of Newton's Universal Law of Gravitation

Consider two objects of masses m_1 and m_2 . We assume that the objects are very small spheres of uniform density and the distance r between their centres is very large compared to the radii of the spheres (Fig.).



The magnitude (F) of the gravitational force of attraction between the objects is directly proportional to m_1m_2 and inversely proportional to r^2

$$\therefore F \propto \frac{(m_1m_2)}{r^2}, \quad \therefore F = \frac{(Gm_1m_2)}{r^2}$$

The above law means that if the mass of one object is doubled, the force between the two objects also doubles. Also, if the distance is doubled, the force decreases by $\frac{1}{4}$.

Q. Why is the constant of gravitation called a universal constant?

- Sol.**
- The value of the constant of gravitation does not change with the nature, mass or the size of the material particles.
 - It does not vary with the distance between the two particles.
 - It is also independent of the nature of the medium between two particles.
- Hence, it is called a universal constant.

The Earth's Gravitational Force

The gravitational force on any object due to the earth is always directed towards the centre of the earth. If the object is on the earth's surface, in the usual notation.

$$F = \frac{(Gm_1m_2)}{r^2}$$



The Earth's Gravitational Acceleration

The acceleration produced in a body due to the earth's gravitational force is called the acceleration due to gravity or the earth's gravitational acceleration and its magnitude is denoted by g . It is directed towards the earth's centre.

Suppose that a body of mass m is released from a distance r from the centre (O) of the earth.

Let M be the mass of the earth. According to Newton's law of gravitation the magnitude of the earth's gravitational force acting on the body is

$$F = G \frac{Mm}{r^2}$$

Where G is the universal constant of gravitation.

The acceleration produced by this force,

$$g = \text{Force/mass} = F/m$$

$$\therefore g = \frac{GM}{r^2}$$

This is the formula for the acceleration due to gravity or the gravitational acceleration due to the earth. This acceleration is directed towards the earth's centre.

If h denotes the altitude, $r = R + h$, where R is the radius of the earth.

$$g = \frac{GM}{(R + h)^2}$$

For a body on the earth's surface, $h = 0$ $g = \frac{GM}{R^2}$

$g = \frac{GM}{r^2}$ for $r \geq R$ (radius of the earth). It depends on the location of the body.

Factors Affecting the Value of g

The value of the acceleration due to gravity, g , changes from place to place on the earth. It also varies with the altitude and depth below the earth's surface.

The factors affecting the value of g are the shape of the earth, altitude and depth below the earth's surface.

(1) The earth is not perfectly spherical. It is somewhat flat at the poles and bulging at the equator.

At the surface of the earth, the value of g is maximum (9.832 m/s^2) at the poles as the polar radius is minimum, It is minimum (9.78 m/s^2) at the equator as the equatorial radius is maximum.

(2) As the height (h) above the earth's surface increases, the value of g decreases. It varies as $1/(R + h)^2$ where R is radius of the earth.

(3) In the interior of the earth, on the average the value of g is less than that at the earth's surface. As the depth below the earth's surface increases the value of g decreases and finally it becomes zero at the centre of the earth.

Free Fall

Whenever an object moves under the influence of gravity alone, it is said to be falling freely.

For a freely falling object, with $u = 0$ and $a = g$,

We have $v = gt$, $s = \frac{1}{2}gt^2$ and $v^2 = 2gs$ (in the usual notation).

Gravitational Potential Energy

The energy stored in a body due to the gravitational force between the body and the earth is called the gravitational potential energy.

The gravitational potential energy of an object at a height h from the earth's surface

$$= -\frac{GMm}{R + H} = -\frac{mgR^2}{R + h}$$



Escape Velocity

When a body is thrown vertically upward from the surface of the earth, the minimum initial velocity of the body for which the body is able to overcome the downward pull by the earth and can escape the earth is called the escape velocity.

$$\text{Escape velocity} = v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

Important Formulae

The total energy of a body revolving the earth = kinetic energy + potential energy

$$= \frac{1}{2}(mv^2) + \left(-\frac{GMm}{R+h}\right)$$

Uniform circular motion of a planet around the Sun:

(1) Centripetal force acting on the planet, $\frac{mv^2}{r}$ gravitational force exerted by the Sun on the planet,

(2) Centripetal acceleration of the planet,

$$a = \frac{v^2}{r} = \frac{\left(\frac{2\pi r}{T}\right)^2}{r} = \frac{4\pi^2 r}{T^2}$$

(3) The period of revolution of the planet,

$$T = \frac{2\pi r}{v} = \frac{2\pi}{\sqrt{GM}} r^{\frac{3}{2}} = \frac{2\pi}{\sqrt{gR^2}} r^{\frac{3}{2}}$$



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PERIODIC CLASSIFICATION OF ELEMENT

In the year 1817 a German scientist Dobereiner suggested that properties of elements are related to their atomic masses.

“Dobereiner made groups of three elements each, having similar chemical properties and called them triads.” He arranged the three elements in a triad in an increasing order of atomic mass and showed that the atomic mass of the middle element was approximately equal to the mean of the atomic masses of the other two elements.

1. HOWEVER, ALL THE KNOWN ELEMENTS COULD NOT BE CLASSIFIED INTO THE DOBEREINER'S TRIADS.

Sr. No.	Triad	Element-1 Actual atomic mass (a)	Element-2		Element-3 Actual atomic mass (c)
			Mean = $\frac{a+c}{2}$	Actual atomic mass	
1.	Li, Na, K	Lithium (Li) 6.9	Sodium $\frac{6.9+39.1}{2} = 23.0$	(Na) 23.0	Potassium (K) 39.1
2.	Ca, Sr, Ba	Calcium (Ca) 40.1	Strontium $\frac{40.1+137.3}{2} = 88.7$	(Sr) 87.6	Barium (Ba) 137.3
3.	Cl, Br, I	Chlorine (Cl) 35.5	Bromine $\frac{35.5+126.9}{2} = 81.2$	(Br) 79.9	Iodine (I) 126.9

Questions for practice

1. Mg (24.3), Ca (40.1), Sr (87.6)
2. S (32.1), Se (79.0), Te (127.6)
3. Be (9.0), Mg (24.3), Ca (40.1)

2. NEWLANDS' LAW OF OCTAVES:

- The English scientist John Newlands In the year 1866 arranged the elements known at that time in an increasing order of their atomic masses.
- It started with the lightest element hydrogen and ended up with thorium.

He found that every eighth had properties similar to those of the first.

“For example, sodium is the eighth element from lithium and both have similar properties. Also, magnesium shows similarity to beryllium and chlorine shows similarity with fluorine.”

“Newlands compared this similarity with the octaves in music. He called the similarity observed in the eighth and the first element as the Law of octaves.”

Musical Note	Do (Sa)	Re (Re)	Mi (Ga)	Fa (Ma)	Sol (Pa)	La (Dha)	Ti (Ni)
Elements	H	Li	Be	B	C	N	O
	F	Na	Mg	Al	Si	P	S
	Cl	K	Ca	Cr	Ti	Mn	Fe
	CO & Ni	Cu	Zn	Y	In	As	Se
	Br	Rb	Sr	Ce & La	Zr		

3. LIMITATION FOUND IN NEWLANDS' OCTAVES:



- This law was found to be applicable only upto calcium.
- Newlands fitted all the known elements in a table of 7 X 8 that is 56 boxes.
- Newlands placed two elements each in some boxes to accommodate all the known elements in the table. For example, Co and Ni, Ce and La.
- Moreover, he placed some elements with different properties under the same note
- Newlands' octaves did not have provision to accommodate the newly discovered elements.
- The properties of the new elements discovered later on did not fit in the Newlands' law of octaves.

4. **MENDELEEV'S PERIODIC TABLE:**

- The Russian scientist Dmitri Mendeleev developed the periodic table of elements during the period 1869 to 1872 A.D.
- Mendeleev's periodic table is the most important step in the classification of elements.
- Mendeleev considered **the atomic mass**, as fundamental property of elements.
- Mendeleev arranged all the 63 known element at that time in an increasing order of their atomic masses.
- Then he transformed this into the periodic table of elements in accordance with the physical and chemical properties of these elements.
- Mendeleev found that the elements with similar physical and chemical properties repeat after a definite interval. On the basis of this finding Mendeleev stated the following **periodic law**.

“Properties of elements are periodic function of their atomic masses.”

Merits of Mendeleev's periodic table:

1. Incorrect mass corrected for example, the previously determined atomic mass of beryllium, 14.09, was changed to the correct value 9.4, and beryllium was placed before boron.
2. Mendeleev kept vacant places in the periodic table for elements not discovered till then
3. Three of these unknown elements were given the names eka-boron, eka-aluminium and eka-silicon from the known neighbors and their atomic masses were indicated as 44, 68 and 72, respectively.

Demerits of Mendeleev's periodic table:

1. Sequence of the elements cobalt (Co) and nickel (Ni)
2. Placement of isotopes in Mendeleev's periodic table.
3. Position of hydrogen

5. **Modern Periodic Law:**

In 1913 A.D. the English scientist Henry Moseley demonstrated, with the help of the experiments done using X-ray tube, that the atomic number (Z) of an element corresponds to the positive charge on the nucleus or the number of the protons in the nucleus of the atom of that element. This revealed that 'atomic number' is a more fundamental property of an element than its atomic mass.

“Properties of elements are a periodic function of their atomic numbers.”

Modern periodic table: long form of the periodic table In the modern periodic table, the elements are arranged in accordance with their atomic number. As a result, most of the drawbacks of Mendeleev's periodic table appear to be removed. However, the ambiguity about the position of hydrogen is not removed even in the modern periodic table. The relation between the atomic number of an element and its electronic configuration is clearly seen in the modern periodic table.

Structure of the Modern Periodic Table

1. 1-7 horizontal rows called Periods
2. 18 vertical columns called Groups
3. There are 118 boxes in the periodic table; it means that there are 118 places for elements in the modern periodic table.
4. Four blocks are there namely s-block, p-block, d-block and f-block



5. Group 1 and 2 are called as **s-block**
6. Group 13-18 are called as **p-block**
7. Group 3-12 are called as d block
8. Lanthanide and actinide series at the bottom called as **f-block**

Periodic trends in the modern periodic table:

1. Atomic size
2. Metallic- Nonmetallic Character
3. Non-Metallic- Nonmetallic Character
4. Electronegativity
5. Electropositivity

Gradation in Halogen Family

The group 17 contains the members of the halogen family. All of them have the general formula X_2 . A gradation is observed in their physical state down the group. Thus, fluorine (F_2) and chlorine (Cl_2) are gases, bromine (Br_2) is a liquid while iodine (I_2) is a solid.



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CHEMICAL REACTION AND EQUATION

Chemical reaction and equations

A chemical reaction is a process in which some substances undergo bond breaking and are transformed into new substances by formation of new bonds.

The substances taking part in chemical reaction are called reactants, whereas the substances formed as a result of a chemical reaction by formation of new bonds are called products.

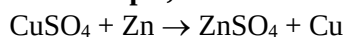
A simple way of representation a chemical' reaction in words is called a 'Word Equation'.

For example,

Aqueous solution of copper sulphate + Zinc dust $\rightarrow$ Aqueous solution of zinc sulphate + Copper

The representation of a chemical reaction in a condensed form using chemical formulae is called as the chemical equation

For example,



Steps in balancing a chemical reaction:

Step 1: Write the chemical equation from the given word equation (Follow this step only when word equation is given).

Step 2: Check whether the equation is balanced or not by comparing the number of atoms of the various elements present on the two sides of the equation.

Step 3: It is convenient to start balancing an equation from the compound which contains the maximum number of atoms. Moreover it is convenient to first consider that element in this compound, which has unequal number of atoms on the two sides.

Step 4: Write down the final balanced equation again.

Types of chemical reactions

When two or more reactants combine in a reaction to form a single product, it is called a **combination reaction**.

The chemical reaction in which two or more products are formed from a single reactant is called a **Decomposition reaction**.

The reaction in which the place of the ion of a less reactive element in a compound is taken by another more reactive element by formation of its own ions, is called **displacement reaction**.

The reaction in which the ions in the reactants are exchanged to form a precipitate are called **double displacement reactions**.

Any process which requires or absorbs heat from its surrounding is called an **endothermic process**.

Any process which releases heat to its surrounding is called an **exothermic process**.

A chemical reaction in which heat is either absorbed or has to be supplied continuously from outside is called **endothermic reaction**.

A chemical reaction in which heat is given away when reactants are transformed into the products is called an **exothermic reaction**.

Factors affecting the rate of a chemical reaction:

1. Nature of the Reactants

Nature or reactivity of reactants influences the rate of a chemical reaction.

2. Size of the Particles of Reactants:

Smaller the size of the reactant particles, higher is the rate of the reaction.

3. Concentration of the reactants:

Rate of a reaction is proportional to the concentration of reactants.

4. Temperature of the Reaction:

Rate of a reaction increases on increasing the temperature.

5. Catalyst

The substance in whose presence the rate of a chemical reaction changes, without causing any chemical change to it, is called a catalyst.

Rate of a reaction is faster when a catalyst is used.

The chemical reaction in which a reactant combines with oxygen or loses hydrogen to form the product is called **oxidation reaction**.

A chemical substance which bring about an oxidation reaction by making oxygen available is called an **oxidant or oxidizing agent**.

The chemical reaction in which a reactant combines with hydrogen or loses oxygen to form the product is called **reduction reaction**.

A chemical substance which bring about a reduction reaction is called a **reductant or reducing agent**.

A chemical reaction in which oxidation and reduction takes place simultaneously is called a **redox reaction**.

Any process causing deterioration or destruction of any material is known as **corrosion**.

Rancidity

It is a condition in which the substance with oil and fats get oxidized when they are exposed to air. A substance is said to be rancid when there is a change in smell, taste and colour.





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EFFECTS OF ELECTRIC CURRENT

Electrical Conductor

A material which has very low electrical resistance is called a good conductor of electricity.

Ex. Silver, copper, aluminium.

Electricity Insulator

A material which has extremely high electrical resistance is called an insulator of electricity.

Ex. Rubber, wood, glass.

Energy Transfer in Electric Circuit

Electric power

Electric power is the electric work done per unit time or electric energy used per unit time. Its SI unit is the watt (W).

Electric power (P) = $\frac{\text{electrical work (W) or electrical energy used}}{\text{time (t)}}$

$$P = VI = I^2R = V^2/R.$$

Mechanical power

Similar to the electric power we can write.

Mechanical power (P) = $\frac{\text{Mechanical work (W)}}{\text{time (t)}}$

$$P = W/t = F \cos \theta / t$$

The watt

In one joule of electric work is done per second, the electric power is 1 watt.

$$1 \text{ watt (W)} = 1 \text{ joule (J)} / 1 \text{ second (s)}$$

Commercial unit of electric energy

The commercial unit of electric energy is the kilowatt-hour (kw.h),

$$1 \text{ kW.h} = 3.6 \times 10^6 \text{ J}$$

It is commonly known as the unit.

The unit of electric power 1W is a very small unit, hence 1000W or 1 kW is used as a unit to measure electric power, in practice. If 1kW power is used for 1 hour, it will mean 1 kW $\times$ 1 h of electric energy is used

$$\begin{aligned} 1 \text{ KW.h} &= 1 \text{ kilowatt-hour} = 1000 \text{ W} \times 3600 \text{ s} \\ &= 3.6 \times 10^6 \text{ W.s} = 3.6 \times 10^6 \text{ J} \end{aligned}$$

Electricity bill

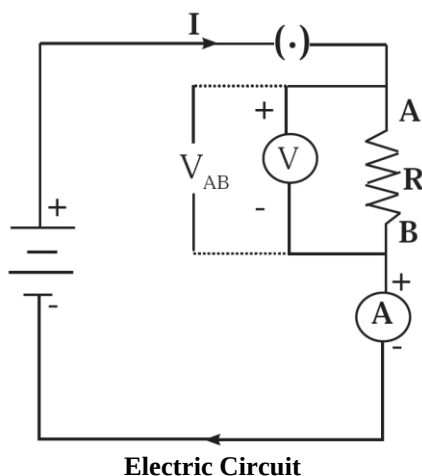
Electricity bill shows the consumption of units (i.e., kWh) and the cost of using electric energy. When 1 kWh electric energy is used, it is termed as 1 unit of energy.

Heating Effect of Electric Current

The production of heat in a resistance due to the electric current flowing through it when it is connected in an electrical circuit, is called the heating effect of electric current.



Mathematical Expression for the Heat Generated in a Metallic Conductor by Electric Current



Fuse

A fuse protects electrical circuits and appliances by stopping the flow of electric current when it exceeds a specified value. For this, it is connected in series with the appliance (or circuit) to be protected.

A fuse is a piece of wire made of an alloy of low melting point (e.g. an alloy of lead and tin). If a current larger than the specified value flows through the fuse, its temperature increases enough to melt it. Hence, the circuit breaks and the appliance is protected from damage.

Magnetic Effect of Electric Current

Passage of electric current through a conductor also produces a magnetic field around it. This effect called magnetic effect of electric current, was discovered by Hans Christian Oersted. The unit of intensity of magnetic field, the oersted, is named after him.

Observation and Conclusion of Oersted's experiment

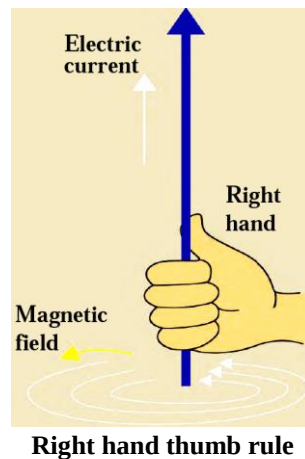
- In Oersted's experiment, when there is no current in the wire, the magnetic needle is at rest along the north-south direction.
- When a current is passed through the wire, the needle is deflected.
- When the current through the wire is increased, the deflection of the needle increases.
- When the current through the wire is stopped the needle comes to rest in its original position along the north-south direction.
- When the current through the wire is reversed, the needle is deflected in the direction opposite to that in the first case.
- When the distance between the magnetic needle and the wire is increased, keeping the current through the wire constant, the deflection of the needle becomes less.

Conclusion

- An electric current produces a magnetic field around it. The moving charge in the conducting wire is a source of magnetic field.
- The direction of the magnetic field produced by the current is the direction in which the north pole of the magnetic needle is deflected.
- At any point near the current-carrying conductor, the magnetic field is perpendicular to (i) the length of the conductor and (ii) the line joining the conductor and the given point.

Right Hand Thumb Rule

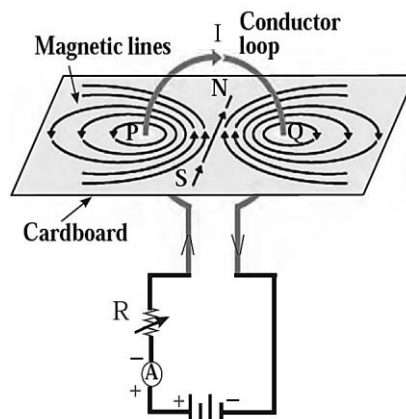
Imagine that you have held a current—carrying straight conductor in your right hand in such a way that your thumb points in the direction of the current. Then turn your fingers around the conductor. The direction of the fingers is the direction of the magnetic lines of force produced by the current.



Right hand thumb rule

Magnetic Field Due to a Current Through a Circular Loop of a Conducting Wire

- The intensity of the magnetic field due to a current through a circular loop of a conducting wire is maximum at the centre of the loop.
- In the case of a circular coil, the magnetic field is proportional to the number of turns of the coil, i.e., if there are n turns in the loop, the magnetic field produced will be n times that produced by a single loop.



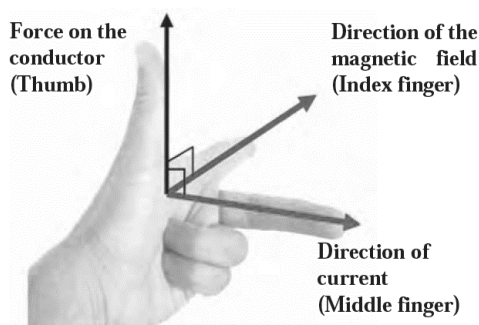
Magnetic field produced by a current through a loop of conducting wire

Magnetic Field due to a Current in a Solenoid

The magnetic field lines due to a current in a solenoid are similar to those in the case of a bar magnet. A current-carrying solenoid behaves like a bar magnet and with it we can magnetise iron and some alloys.

Fleming's Left Hand Rule

The left hand thumb, index finger, and the middle finger are stretched so as to be perpendicular to each other. If the index finger is in the direction of the magnetic field, and the middle finger points in the direction of the current, then the direction of the thumb is the direction of the force on the conductor.



Fleming's left hand rule

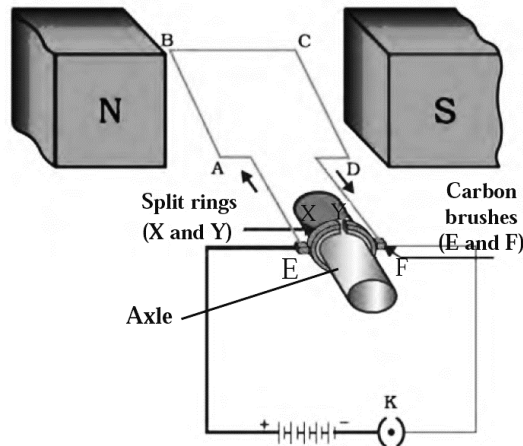


Electric Motor

An electric motor is a device that converts electric energy into mechanical energy. It operates on the principle that a current carrying conductor in a magnetic field experiences a force.

Construction and Working of Electric Motor

Below figure shows the construction of an electric motor



Electric motor: Principle and Working

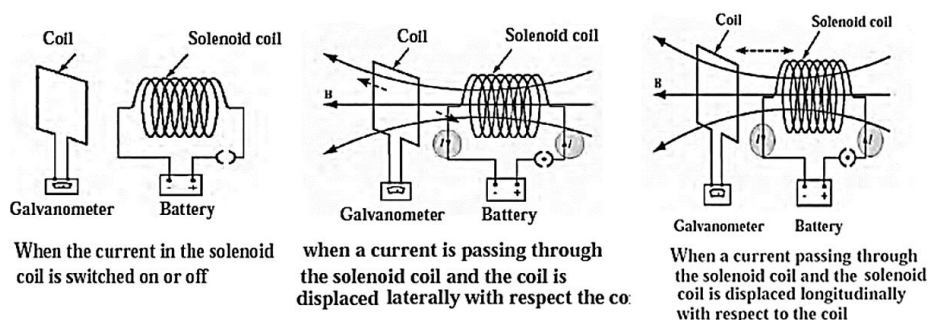
- In this case, a rectangular loop ABCD of copper wire with a resistive coating is placed between the north and south poles of a strong magnet, such as a horseshoe magnet, with the branches AB and CD perpendicular to the magnetic field direction.
- The loop's ends are connected and tightly fitted on the axle. X and Y's outer conducting surfaces are in contact with two stationary carbon brushes, E and F.

Working

- When the circuit is completed with a plug key or switch, the current flows in the direction $E \rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow F$.
- Because the magnetic field is directed from north pole to south pole, the force on AB is downward and the force on CD is upward, according to Fleming's left hand rule. As a result, AB moves downward and CD moves upward. These forces have the same magnitude but opposite directions.
- Therefore, as observed from the side AD, the loop ABCD and the axle start rotating in anticlockwise direction.
- After half a rotation, X and Y make contact with brushes F and E, causing current to flow in the direction EDCBAF. As a result, the force on CD is downward and the force on AB is upward. As a result, the loop and axle continue to rotate anticlockwise.
- After every half rotation, the current in the loop is reversed and the loop and the axle continue to rotate in anticlockwise direction.
- When the current is switched off, the loop stops rotating after some time.

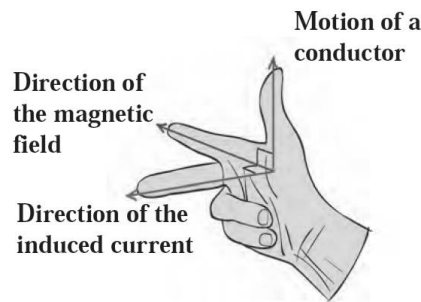
Faraday's Law of Induction

Whenever the number of magnetic lines of force passing through a coil changes, a current is induced in the coil. The current produced in the coil is called the induced current.





Fleming's Right Hand Rule



Stretch the thumb, the index finger and the middle finger of the right hand in such a way that they are perpendicular to each other. In this position, the thumb indicates the direction of the motion of the conductor, the index finger the direction of the magnetic field and the middle finger shows the direction of the induced current.

Electric Generator

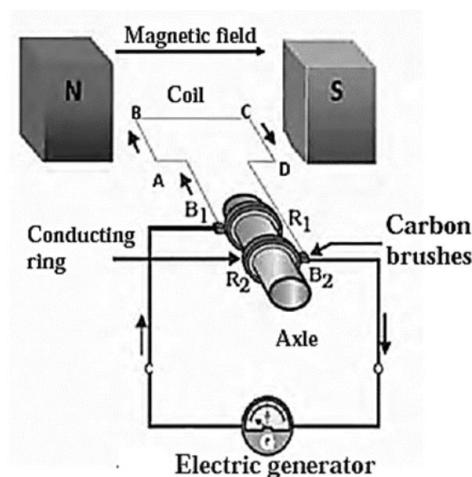
An electric device which converts mechanical energy into electric energy is called an electric generator.

- Its working is based on electromagnetic induction.
- When the coil of an electric generator rotates in a magnetic field, a current is induced in the coil. This induced current then flows in the circuit connected to the coil.
- A generator producing a direct current is called a DC generator and that producing an alternating current is called an AC generator.

Construction and Working of an Electric Generator (AC)

Construction

Below fig shows a copper wire coil ABCD, kept between the two pole pieces of a magnet.



- The two ends of the coil are connected to the conducting rings R_1 and R_2 via carbon brushes.
- Both the rings are fixed to the axle, but there is a resistive coating in between the ring and the axle.
- The axle is rotated with the help of a machine from outside. Because of this, the coil ABCD starts rotating.
- The stationary carbon brushes B_1 and B_2 are connected to a galvanometer, which shows the direction of current in the circuit.
- Upon rotating the axle, the branch AB goes up and the branch CD goes down (i.e. the coil ABCD rotates clockwise).



Working

According to Fleming's right hand rule, electric current is produced in the branches AB and CD in the direction. $A \rightarrow B$ and $C \rightarrow D$. Thus, the current flows in the direction $A \rightarrow B \rightarrow C \rightarrow D$ (as shown by arrows in the figure).

In the external circuit, the current flows from B_2 to B_1 through the galvanometer.

If instead of one loop coil, a coil consisting of several turns is used, the current of magnitude several times flows.

After half rotation, the branch AB takes the place of branch CD and the branch CD takes the position of the branch AB.

Therefore, the induced current goes as $D \rightarrow C \rightarrow B \rightarrow A$.



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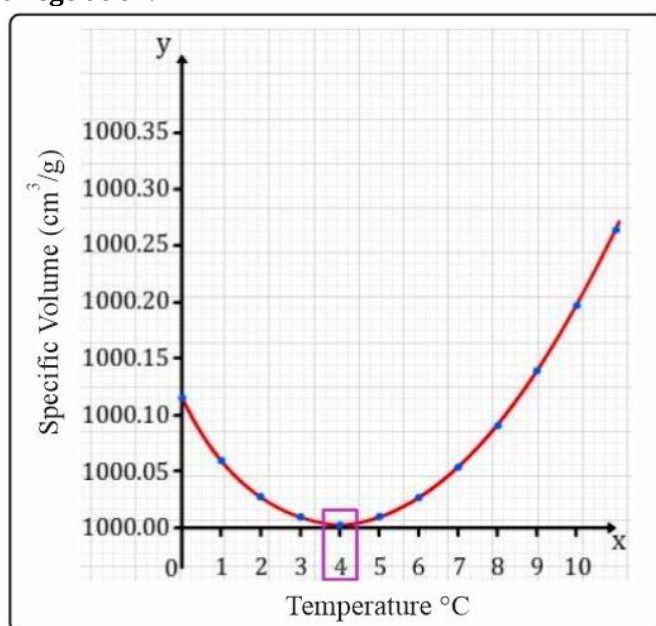
HEAT

Heat

The amount of heat energy absorbed at constant temperature by unit mass of a solid to convert into liquid phase is called the specific **latent heat of fusion**.

The amount of heat energy absorbed at constant temperature by unit mass of a liquid to convert into gaseous phase is called the specific **latent heat of vaporization**.

The phenomenon in which the ice converts to liquid due to applied pressure and then re-converts to ice once the pressure is removed is called **regelation**.



If we heat water from 0°C upto 4°C, it contracts instead of expanding. At 4°C its volume is minimum (due to contraction). If the water is heated further, it expands and its volume increases. The behaviour of water between its temperature from 0°C to 4°C is called **anomalous behaviour of water**.

Suppose unsaturated air at a certain temperature is taken and its temperature is decreased, a temperature is reached at which the air becomes saturated with vapor. This temperature is called the dew point temperature.

The ratio of actual mass vapor content in the air for a given volume and temperature to that required to make the air saturated with vapor at that temperature is called the relative humidity.

% **Relative humidity**: -

$$\frac{\text{Actual mass of water vapor content in the air in a given volume}}{\text{mass of vapor needed to make the air saturated in that volume}} \times 100$$

The relative humidity at the dew point is 100%. If the relative humidity is more than 60%, we feel that the air is humid. If the relative humidity is less than 60%, **we feel that the air is dry**.

The amount of heat necessary to raise temperature of 1 g of water by 1° C from 14.5°C to 15.5°C is called one called **heat**.



If we heat 1 kg of water by 1°C in different temperature range than 14.5°C to 15.5°C , the amount of heat required will be slightly different. It is, therefore, essential to define a specific temperature range while defining the unit of heat. Calorie and Joule are related by following relation:

$1\text{ cal} = 4.18\text{ Joule}$

The amount of heat energy required to raise the temperature of a unit mass of an object by 1°C is called the **specific heat of that object**.

Heat energy lost by the hot object is equal to the heat energy gained by the cold object. This is called as '**Principle of heat exchange**'.





2024-25

REFRACTION OF LIGHT

Reflection of light

When light is incident on the surface of an object, in general, it is deflected in different directions. This process is called reflection of light.

Laws of reflection of light

- The incident ray and the reflected ray of light are on the opposite sides of the normal to the reflecting surface at the point of incidence and all the three are in the same plane.
- The angle of incidence and the angle of reflection are equal in measure.

Refraction of light

Light changes its direction when going from one transparent medium to another transparent medium. This is called the refraction of light.

Refraction occurs as the velocity of light is different in different media.

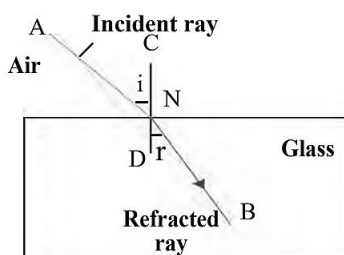
Laws of refraction of light:

- The incident ray and the refracted ray are on the opposite sides of the normal to the surface at the point of incidence and all the three, i.e. the incident ray, the refracted ray and the normal are in the same plane.
- For a given pair of media, the ratio of the sine of the angle of incidence ($\sin i$) to the sine of the angle of refraction ($\sin r$) is constant (Snell's law). This constant is called the refractive index of the second medium with respect to the first medium. Here, i is the angle of incidence and r is the angle of refraction.

Refractive index

If i is the angle of incidence and r is the angle of refraction, then

$$\frac{\sin i}{\sin r} = \text{constant (Snell's law)}$$

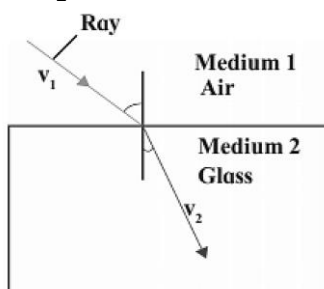


Light ray entering a glass slab from air

This constant is called the refractive index of the second medium with respect to the first medium, and is denoted by ${}_2n_1$.

Thun, ${}_2n_1 = \frac{\sin i}{\sin r}$ Also, if v_1 , is (the magnitude of) the velocity of light in the first medium and v_2 is (the magnitude of) the

velocity of light in the second medium then ${}_2n_1 = \frac{v_1}{v_2}$.





Similarly, the refractive index of the first medium with respect to the second medium is given by ${}_1n_2 = \frac{v_2}{v_1}$.

If the first medium is vacuum, ${}_2n_1$ is considered with respect to vacuum. It is called the absolute refractive index of the medium 2 and is denoted by n .

Refractive index depends on the velocity of light in the medium.

The absolute refractive index of air is 1.0003. This shows that for almost all practical purposes, the speed of light in air is very nearly the same as that in vacuum.

$$\frac{\sin r}{\sin i} = \text{constant} = n$$

n is called the refractive index of the second medium with respect to the first medium. This second law is also called Snell's law. A ray incident along the normal ($i = 0$) goes forward in the same direction ($r = 0$).

Absolute refractive indices of different media

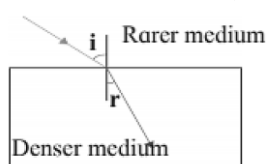
Substance	Refractive index	Substance	Refractive index	Substance	Refractive index
Air	1.0003	Fused Quartz	1.46	Carbon disulphide	1.63
Ice	1.31	Turpentine oil	1.47	Dense flint glass	1.66
Water	1.33	Benzene	1.50	Ruby	1.76
Alcohol	1.36	Crown glass	1.52	Sapphire	1.76
Kerosin	1.39	Rock salt	1.54	Diamond	2.42

Absolute refractive indices of some media

Behaviour of a ray of light in refraction :

When a light ray passes from a rarer medium to a denser a medium

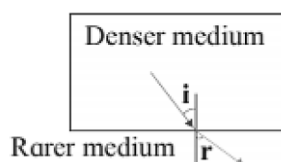
- When a light ray passes from a rarer medium to a denser a medium, it bends towards the normal at the point of incidence



Here, i is greater than r , and ${}_2n_1$ is greater than 1. The greater the value of ${}_2n_1$ the greater is the bending towards the normal.

When a light ray passes from a denser medium to a rarer medium

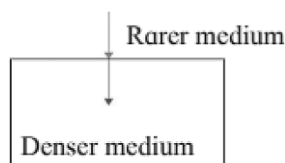
- When a light ray passes from a denser medium to a rarer medium, it bends away from the normal at the point of incidence.



Here, r is greater than i , and ${}_2n_1$ is less than 1. The greater the value of ${}_2n_1$ the less is the bending away from the normal.

When a light ray is incident normally at the boundary between two media

When a light ray is incident normally at the boundary between two media, it does not change its direction and hence does not get refracted.





If a ray of light is incident normal to the interface between any two media (whether passing from an optically rarer medium to an optically denser medium or from an optically denser medium to an optically rarer medium), the angle of incidence is zero and so also the angle of refraction. Here, the light goes ahead in the same direction.

Refraction of light passing through a glass slab and Refraction of light passing through transparent plastic slab.

Similarity

The emergent ray is parallel to the incident ray, but it is displaced slightly sideways.

Difference

For a given angle of incidence, the extent of refraction (bending) is different (in general, less) for a transparent plastic slab relative to the glass slab.

Q. Will light travel through a glass slab with the same velocity as it travels in air?

Sol. No, Light won't travel through a glass slab with the same velocity as it travels in air.

Q. Will the velocity of light be same in all media?

Sol. No, because of the interaction between light and particles of the medium (atoms, molecules, etc.) the velocity of light will not be same.

Twinkling of a star and atmospheric refraction

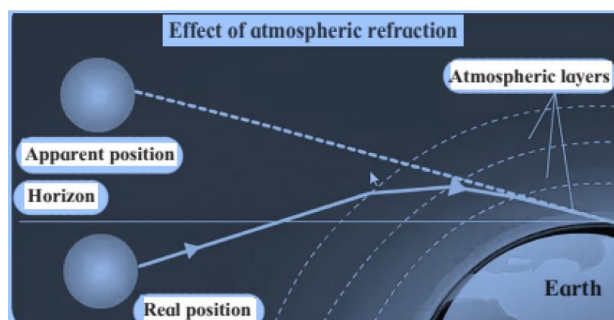
- As a star is far away from the earth, it appears as a point source of light.
- When Starlight enters the earth's atmosphere, it undergoes refraction continuously in the medium with gradually varying refractive index.
- The bending of starlight occurs towards the normal as it passes from the optically rarer part of the medium to the optically denser part.
- Hence, when a star is observed near the horizon, its apparent position is slightly higher than the actual position. Further, the apparent position varies with time as the medium is not stationary.
- Also, there is fluctuation in the brightness of a star when observed from the earth. This is called twinkling of a star.
- The planets are relatively closer to the earth. Hence they appear as a collection of a large number of point sources of light. The net fluctuation in the brightness of a planet, therefore, turns out to be zero. Also, there is no change in the average position of a planet. Hence, planets do not twinkle.

Advanced sunrise and delayed sunset

The advanced sunrise and delayed sunset are also the result of atmospheric refraction.

(1) The sunrise (the appearance of the sun above the horizon) is advanced due to atmospheric refraction of sunlight. An observer on the earth sees the sun two minutes before the sun reaches the horizon.

- A ray of sunlight entering the earth's atmosphere follows a curved path due to atmospheric refraction before reaching the earth.
- This happens due to a gradual variation in the refractive index of the atmosphere.
- For the observer on the earth, the apparent position of the sun is slightly higher than the actual position.
- Hence, the sun is seen before the sun reaches the horizon.





- (2) Increased atmospheric refraction of sunlight occurs also at the sunset i.e. when the sun disappearing below the horizon. In this case, the observer on the earth continues to see the setting sun for two minutes after the sun has dipped below the horizon, thus delaying the sunset.

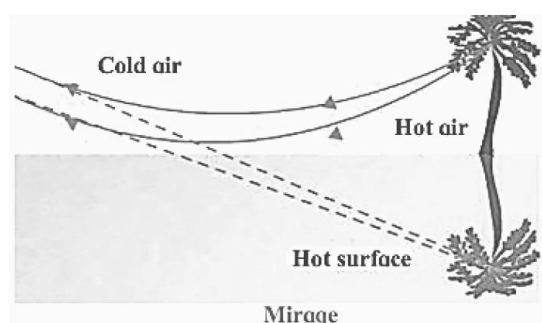
The advanced sunrise and delayed sunset increases the duration of day by four minutes

Local atmospheric conditions affect refraction of light

Mirage, Shaking of objects appear beyond and above a holi fire are examples of the effect of atmospheric refraction.

A mirage

- Due to the changes in refraction of light, the light rays coming from a distant object appear to be coming from the image of the object inside the ground. This is called a mirage.
- In summer, when the earth's surface is heated by the sun, the temperature of air increases. This produces a layer of hot air of lower density (mass per unit volume) and lower refractive index at the surface.
- Results in hot air works as an optically rarer medium relative to cool air.
- When the temperature changes rapidly in the vertical direction, as refraction of light takes place, the angle of refraction changes continuously.
- The rays of light from the top of an object such as a car or tree cross the rays from the bottom of the object on their way to the observer's eye.
- Hence, an inverted image is formed below the object's true position and downward towards the surface in the direction of air at higher temperature. In this case, some rays of light bend back up into the denser air above.
- Mirage produces an impression of water near the hot ground,



Dispersion of light

- The process of separation of light into its component colours while passing through a medium is called dispersion of light.
- The band of coloured components of a light beam is called its spectrum.
- The formation of a rainbow is due to refraction, dispersion, internal reflection and again refraction of sunlight by water droplets under appropriate conditions.
- Our eyes are sensitive to electromagnetic radiation of wavelength in the range 400 nm to 700 nm. [1nm=10⁻⁹m; nm = nanometer]
- Wavelength of red light is close to 700 nm and that of violet light is close to 400 nm.

Q. How does the dispersion of white light take place when it passes through a glass prism?

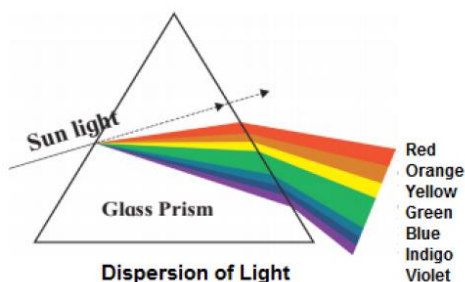
Sol. When rays of light are incident on a prism, they are refracted twice, while travelling from air to glass and then from glass to air. Even when the incident rays are directed away from the base of the prism, the emergent rays bend towards the base of the prism, as the prism is triangular. Thus, the rays are deviated as they pass through the prism.

The refractive index of glass is different for different colours. Therefore, the rays corresponding to different colours are deviated to different extents.

White light is a mixture of seven colours : violet, indigo, blue, green, yellow, orange and red. Hence, when white light is incident on a prism, a spectrum of seven colours is obtained.



The refractive index of glass is maximum for violet light and minimum for red light. Hence, violet light is deviated the most and red light is deviated the least. The deviation of rays corresponding to other colours is intermediate. In this manner, the dispersion of light takes place when it passes through a glass prism.



Partial and total internal reflection

When light travels from one medium to another, part of incident light comes back into the first medium. This is called partial reflection. Remaining part is refracted.

When light travels obliquely from a denser medium to a rarer medium, the angle of refraction r is greater than the angle of incidence i . Also, the ratio $\sin i / \sin r$ is constant.

Critical angle

For a particular value of i , r becomes 90° . This particular angle of incidence is called critical angle

For $i > \text{critical angle}$, as r cannot be greater than 90° , light is totally reflected in the denser medium. This is called total internal reflection.

Here, ${}_2n_1 = \frac{\sin i}{\sin 90^\circ} = \sin i$, where i is the critical angle.



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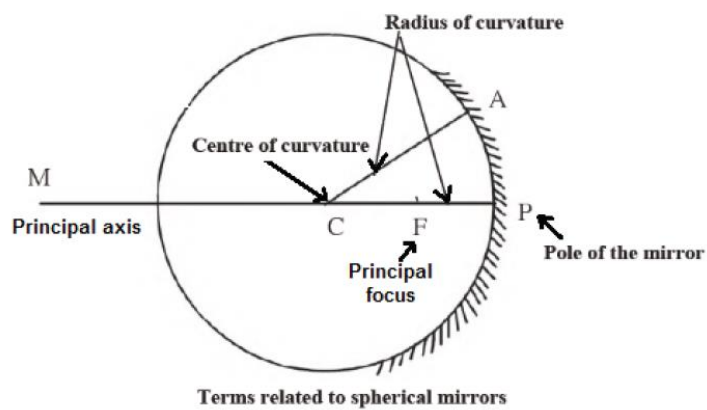
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2024-25

LENSES

Terms related to spherical mirrors



Construction of concave mirrors

The given part of a hollow spherical glass can be converted into a concave mirror by

- (i) Polishing (silvering) its inner side (inner surface or concave surface) to make it reflecting (front surface silvered concave mirror)
or
- (ii) Coating its outer side with a thin layer of silver and painting it with red colour to protect the silver coating.

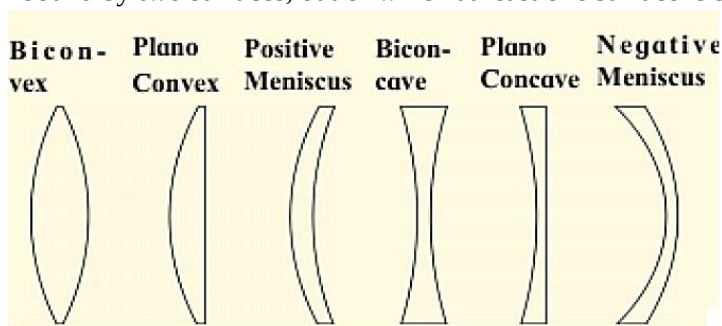
Construction of convex mirrors

The given part of a hollow spherical glass can be converted into a convex mirror by

- (i) Polishing — (silvering) its outer side (outer surface or convex surface) to make it reflecting. (front surface silvered convex mirror)
or
- (ii) Coating its inner side with a thin layer of silver and painting it with red colour to protect the silver coating.

Lens

A lens is a transparent material bound by two surfaces, out of which at least one surface is spherical.



Types of lenses

Difference between lens and mirror

- A mirror has one reflecting surface. By reflection of light, it forms an image of the object placed in front of it. A mirror is not transparent.
- A lens has two surfaces that form an image by refraction of light. A lens is transparent.



Terms of lenses :

i. Centre of curvature (C) of a lens

The centres of the spheres whose parts form the surfaces of a lens are called the centres of curvature of the lens. A lens has two centres of curvature C_1 and C_2 for its two spherical surfaces.

ii. Radii of curvature (R_1 , R_2) of a lens

The radii of the spheres whose parts form surfaces of a lens are called the radii of curvature of the lens.

iii. Principal axis of a lens

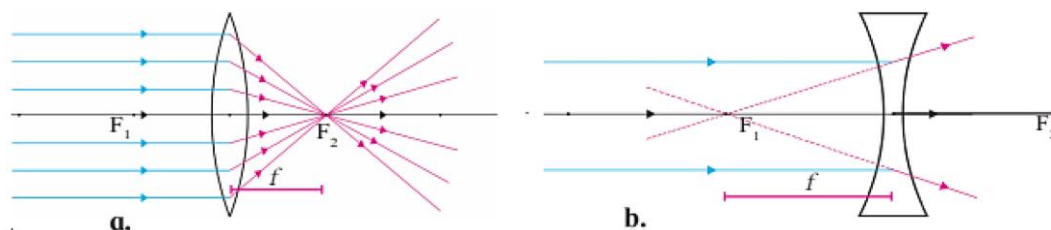
The imaginary straight line passing through the two centres of curvature of a lens is called the principal axis of the lens.

iv. Optical centre (O) of a lens

The point inside a lens on the principal axis, through which light rays pass without changing their path is called the optical centre (O) of the lens.

v. Principal focus (F) of a lens

When light rays parallel to the principal axis are incident on a convex lens, they converge at a point on the principal axis. This point is called the principal focus (F) of the convex lens. Light rays travelling parallel to the principal axis of a concave lens diverge after refraction in such a way that they appear to be coming out of a point on the principal axis. This point is called the principal focus of the concave lens. A lens has two principal foci F_1 , and F_2 .



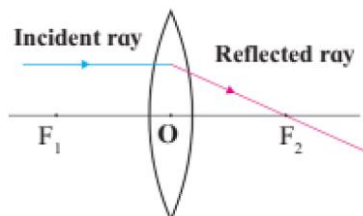
Principle focus of lens

Convex lens

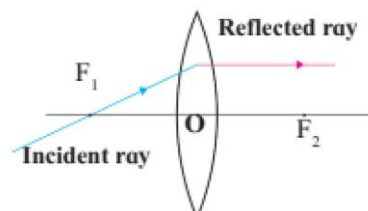
- A lens having both spherical surfaces puffed up outwards is called a convex lens or double convex lens or biconvex lens.
- A lens having one surface plane and the other (spherical surface) bulging outward is called a planoconvex lens.
- A convex lens is thicker in the middle than at the edges.
- It is a converging lens.
- The concavo-convex lens has one spherical surface concave and the other convex such that it behaves as a convex lens.

Rules for obtaining an image formed by a convex lens

- (i) When the incident ray is parallel to the principal axis, the refracted ray passes through the principal focus.

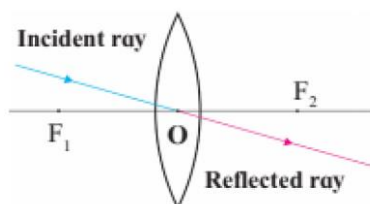


- (ii) When the incident ray passes through the principal focus, the refracted ray is parallel to the principal axis.

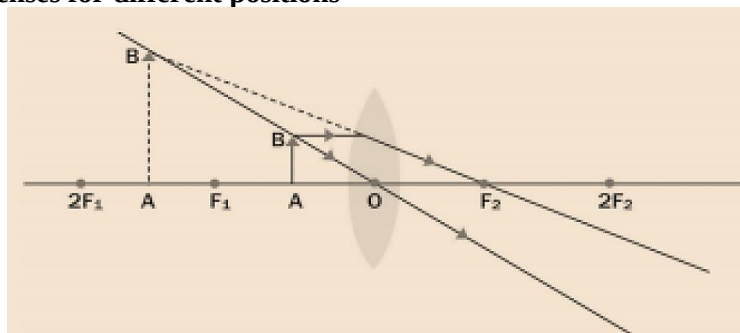




(iii) When the incident ray (ray of light) passes through the optical centre of the lens, it passes without changing its direction.



Images formed by convex lenses for different positions



S. No.	Position of the object	Position of the image	Size of the image	Nature of the image
1	At infinity	At focus F_2	Point image	Real and inverted
2	Beyond $2F_1$	Between F_2 and $2F_2$	Smaller	Real and inverted
3	At $2F_1$	At $2F_2$	Same size	Real and inverted
4	Between F_1 and $2F_1$	Beyond $2F_2$	Larger	Real and inverted
5	At focus F_1	At infinity	Very large	Real and inverted
6	Between F_1 and O	On the same side of the lens as the object	Very large	Virtual and erect

Images formed by convex lenses for different positions of the object

Concave lens

- A lens having both spherical surfaces curved inwards is called a concave lens or double concave lens or biconcave lens.
- A lens having one surface plane and the other (spherical surface) curved inwards is called a plano-concave lens.
- A concave lens is thicker at the edges than in the middle.
- It is a diverging lens.
- The convexo-concave lens has one spherical surface convex and the other concave such that it behaves as a concave lens.

Rules for obtaining an image formed by a concave lens

- When the incident ray is parallel to the principal axis, the refracted ray, when extended backwards, passes through the principal focus.
- When the incident ray is directed towards the principal focus F_2 , the refracted ray is parallel to the principal axis.
- When the incident ray passes through the optical centre of the lens, it passes without changing its direction.

Characteristics of an image formed by a concave lens

The image formed by a concave lens is always virtual, erect and smaller than the object. It is on the same side of the lens as the object.

Generally, it is formed between the optical centre of the lens and the principal focus F_1 . If the object is at infinity, the image is a point image formed at F_1 .

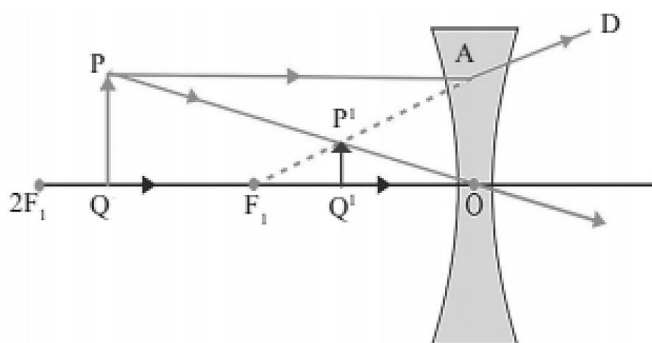


Image formation by a concave lens

- (i) If the object is at infinity, the image is formed at the focus of the lens, on the same side of the lens as the object. It is virtual, erect and much smaller than the object (point image).
- (ii) If the object is at any finite distance from the lens, the image is formed on the same side of the lens as the object and between the focus and the optical centre of the lens. It is virtual, erect and smaller than the object. The image distance is less than the object distance.

Sr. No.	Position of the object	Position of the image	Size of the image	Nature of the image
1	At infinity	On the first focus F_1	Point image	Virtual and erect
2	Anywhere between optical centre O and infinity	Between optical centre and focus F_1	Small	Virtual and erect

Images formed by concave lenses

Working of a lens

The working of a lens is similar to that of a triangular prism. When a ray of light passes through a lens, it is refracted twice,

- When entering the lens and
- when emerging from the lens.

There is a change in its direction of propagation every time and as both the changes occur in the same sense, the direction of propagation of the emergent ray is different from that of the incident ray.

Q. What are real and Virtual images? How will you find out whether an image is real or virtual? Can a virtual image be obtained on a screen?

Sol.

- An image formed by convergence of reflected or refracted rays of light at a point is called a real image.
- An image formed at a point from which the reflected or refracted rays of light appear to diverge is called a virtual image.
- A real image can be obtained on a screen. A virtual image cannot be obtained on a screen. This is how we can find out whether an image is real or virtual.

Sign convention for a lens

In this case, the optical centre (O) of the lens is taken as the origin and the principal axis of the lens is taken as X-axis of the coordinate system. The sign conventions regarding the measurement of distances parallel to the principal axis and those perpendicular to the principal axis are the same as for a spherical mirror. Hence, the focal length of a convex lens is positive and that of a concave lens is negative.

Q. What is the Cartesian sign convention used for spherical mirrors?

Sol. According to the Cartesian sign convention, the pole (P) of a spherical mirror is taken as the origin and the principal axis is taken as X-axis of the coordinate system.

- (1) The object is always placed on the left of the mirror.
- (2) All distances parallel to the principal axis are measured from the pole of the mirror.
- (3) All distances measured to the right of the origin (pole) are taken as positive while distances measured to the left of the origin (pole) are taken as negative.
- (4) Distances measured perpendicular to and above the principal axis are taken as positive.
- (5) Distances measured perpendicular to and below the principal axis are taken as negative.
- (6) The focal length of a convex mirror is positive while that of a concave mirror is negative.



Lens formula

The relationship between the object distance (u), image distance (v) and focal length (f) of a lens is called the lens formula and is given as.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Magnification by a lens

The magnification (M) produced by a lens

$$= \frac{\text{height of the image } h_2}{\text{height of the object } h_1} = \frac{v}{u}$$

Magnification is positive for a virtual image and negative for a real image.

Power of a lens

The power (P) of a lens

$$= \frac{1}{\text{focal length (f) of the lens}}$$

Its SI unit is the dioptre (D)., If f = 1 metre, P = 1 dioptre.

iii. Combination of lenses

If two lenses with focal lengths f_1 and f_2 are kept in contact with each other, the effective focal length of the combination, f, is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

The effective power of the combination of the lenses, P, is given by $P = P_1 + P_2$ where

$$P = \frac{1}{f}, P_1 = \frac{1}{f_1}, P_2 = \frac{1}{f_2}$$

Human Eye

Cornea

The cornea is a thin and transparent cover (membrane) on the human eye through which light enters the eye. Maximum refraction of light rays entering the eye occurs at the cornea.

Iris

The iris is a dark fleshy screen (muscular diaphragm) behind the cornea in the human eye. Its colours are different for different people.

Pupil

The pupil is a small circular opening of changing diameter at the centre of the iris in the human eye.

The pupil in the human eye is useful for controlling and regulating the amount of light entering the eye. The pupil contracts in the presence of too much light and dilates when light is insufficient, thus changing the amount of light entering the eye.

Adaptation

The tendency of the pupil in the human eye to adjust the opening for light, depending on the intensity of incident light, to control and regulate the amount of light entering the eye is called adaptation.

Eye Lens

A transparent biconvex crystalline lens is located just behind the pupil in the human eye. The eye lens provides small adjustment of focal length to form a real and inverted sharp image on the retina.

Shape and the size of the human eyeball

The human eyeball is approximately spherical in shape with a diameter of about 2.4cm.



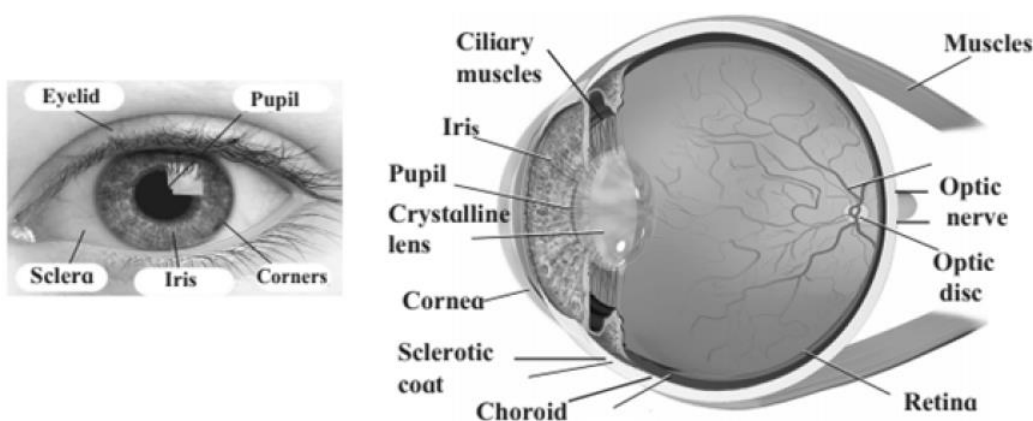
Retina

The retina is a light sensitive screen consisting of a delicate membrane with a large number of light-sensitive cells. Due to these cells, we get information about the brightness or dimness of the object as well as the colour of the object.

When light falls on the retina, light sensitive cells of the retina are activated. They generate electrical signals which are passed by optic nerves to the brain. The brain interprets the signals and processes the information such that we perceive the object as it is.

Ciliary muscles

The muscles which hold the eye lens in its position, and bring about changes in the shape (curvature) of the eye lens, and hence of focal length are known as ciliary muscles.



Construction of Human Eye

Persistence of vision

The image of an object remains imprinted on our retina for $\approx 1/16^{\text{th}}$ of a second after the object is removed from the sight. The sensation on the retina persists for a while. This is called persistence of vision.

Power of accommodation of the eye

The ability of the eye lens to adjust its focal length is called the power of accommodation of the eye.

The minimum distance of distinct vision and the near point

The minimum distance from the normal eye at which an object is clearly visible without stress on the eye is called the minimum distance of distinct vision. It is 25 cm for the normal human eye.

The position of the object at the minimum distance of distinct vision is called the near point of the eye. For a normal human eye, the near point is at 25 cm from the eye.

Defects of vision and their correction :

Problems of vision are related to

- (i) weakening of ciliary muscles
- (ii) change in the size of the eyeball
- (iii) irregularities on the surface of cornea
- (iv) formation of a membrane over the eye lens.

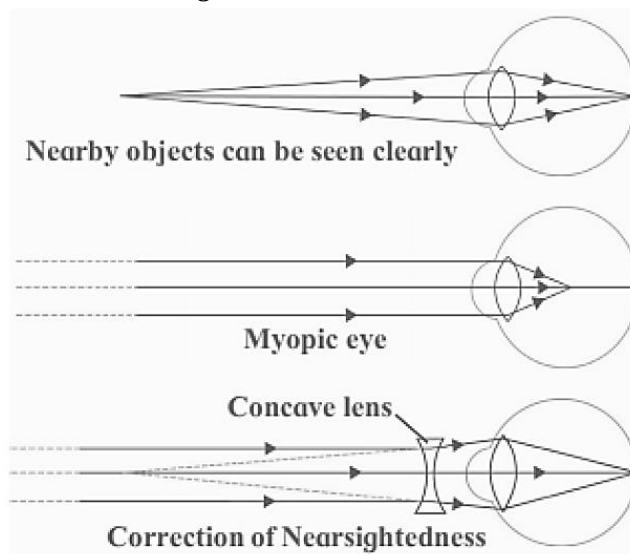
Myopia or Near sightedness

Myopia or near sightedness is the defect of vision in which a human eye can see nearby objects distinctly but is unable to see distant objects clearly. In this case, the image of a distant object is formed in front of the retina.

Myopia is corrected using a suitable concave lens. Light rays are diverged by the concave lens before they strike the eye lens.



A concave lens of proper focal length is chosen to produce the required divergence. Hence, after the converging action of the eye lens, the image is formed on the retina. [Fig]



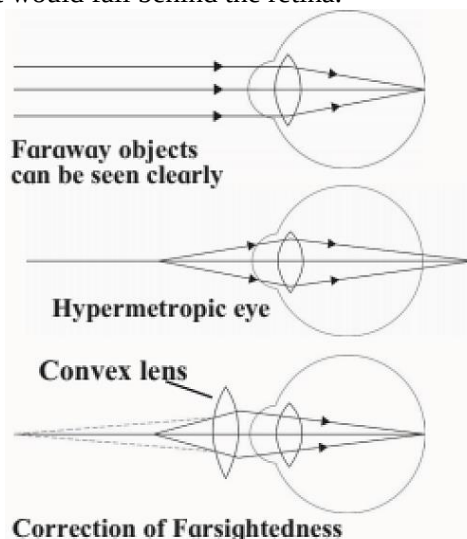
Possible reasons of myopia

- (1) The curvature of the cornea and the eye lens increases. The muscles near the lens cannot relax so that the converging power of the lens remains large.
- (2) The distance between the eye lens and the retina increases as the eyeball elongates.

Hypermetropia or Farsightedness

Hypermetropia or farsightedness is the defect of vision in which a human eye can see distant objects distinctly but is unable to see nearby objects clearly.

In this case, the image of a nearby object would fall behind the retina.



Possible reasons of hypermetropia

- (1) Curvature of the cornea and the eye lens decreases. Hence, the converging power of the eye lens becomes less.
- (2) The distance between the eye lens and retina decreases (relative to the normal eye) and the focal length of the eye lens becomes very large due to the flattening of the eyeball.

Presbyopia (also called old age hypermetropia)

Presbyopia is the defect of vision in which aged people find it difficult to see the nearby objects comfortably and clearly without spectacles.



This defect is corrected using a convex lens of appropriate power. The lens converges light rays before they fall on the eye lens such that the action of the eye lens forms the image on the retina.

Reason of presbyopia

The power of accommodation of eye usually decreases with ageing. The muscles near the eye lens lose their ability to change the focal length of the lens. Therefore, the near point of the eye lens shifts farther from the eye.

Bifocal lens

A bifocal lens is a lens of which the upper part is a concave lens to correct myopia and the lower part is a convex lens to correct hypermetropia.

- A person suffering from myopia as well as hypermetropia, uses a bifocal lens. Nowadays, the defects of vision such as myopia and hypermetropia can be corrected using contact lenses or by laser surgery.

Camera and Human Eye

Cameras, in general, have various shapes and sizes. Some cameras are much bigger than the human eye while some are smaller than the human eye. Here we shall consider a simple camera.

Similarities

- In the case of a camera as well as the human eye, it is possible to control the amount of incoming light with the help of a diaphragm and an aperture.
- Both use a convex lens for focusing.
- The photographic film in a camera is coated with a photosensitive material. The retina in the eye consists of a large number of light sensitive cells.
- The photographic film in a camera is processed using chemicals and then prints (photographs) can be obtained using the appropriate paper.
- In the human eye, the electrical signals generated by light sensitive cells are passed by optic nerves to the brain which interprets them.

Differences

- Cameras come in a variety of sizes and shapes unlike the human eye.
- Unlike the human eye, a wide variation in exposure time is possible in the case of cameras.
- The human eye is sensitive in the visible region (red to violet) of the electromagnetic spectrum, while a much wider range of the electromagnetic spectrum can be covered with cameras designed for specific purposes.
- In comparison with the human eye, a wider view and range can be covered by a camera.
- In comparison with the human eye, a wider intensity (of light) range can be covered with a camera.
- The retina is indispensable in the human eye, while cameras without a photographic film have been designed with the help of photo sensitive materials and are in current use.
- With advances in technology, improved cameras are designed all the time, and the list of differences between the human eye and a camera in general would be practically endless.

Uses of lenses

Uses of a convex lens

Simple microscope, compound microscope, telescope, camera, projector, spectrometer, spectacles, etc., make use of one or more convex lenses. A simple microscope is used by watch repairers, jewellers, etc. A compound microscope is used to observe bacteria, cells, microorganisms, etc.

A telescope is used to observe distant terrestrial objects or astronomical objects like planets, stars and comets.

Uses of a concave lens

A concave lens is used in spectacles to correct myopia. It is also used in optical instruments.

Uses of a concave lens

A concave lens is used in spectacles to correct myopia. It is also used in optical instruments.



Construction of a compound microscope

- (1) A compound microscope consists of a metal tube fitted with two convex lenses at the two ends. These lenses are called the objective lens (the lens directed towards the object) and the eyepiece (the lens directed towards the eye). Both the lenses are small in size, but the cross section of the objective lens is less than that of the eyepiece. The objective lens has a short focal length. The focal length of the eyepiece is more than that of the objective lens.
- (2) The metal tube is mounted on a stand. The principal axes of the objective lens and the eyepiece are along the same line. The distance between the object and the objective lens can be changed with a screw. It is possible to change the distance between the objective lens and the eyepiece.

Working

- (1) The object to be observed is illuminated and placed in front of the objective lens, slightly beyond the focal length of the objective lens. It's real, inverted and enlarged image is formed by the objective lens on the other side.
- (2) This intermediate image lies within the focal length of the eyepiece. It serves as an object for the eyepiece. The eyepiece works as a simple microscope. The final image is virtual, highly enlarged and inverted with respect to the original object. It can be formed at the minimum distance of distinct vision from the eyepiece. The final image is observed by keeping the eye close to the eyepiece.

Uses of a compound microscope

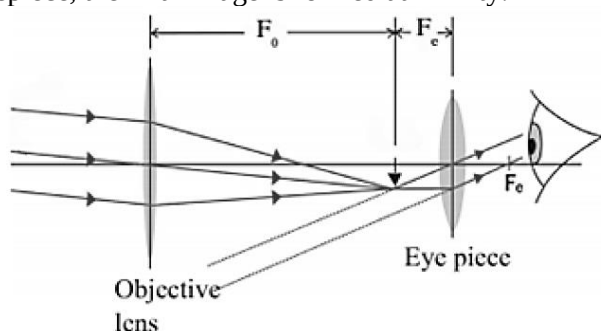
- It is used to observe blood corpuscles, plant and animal cells, microorganisms like bacteria, etc.
- It is used in a pathological laboratory to observe blood, urine, etc.
- It is a part of a travelling microscope used for measurement of very small distance.

Construction of a refracting telescope (Astronomical telescope)

- It consists of two convex lenses called the objective lens (directed towards the object) and the eyepiece (directed towards the eye).
- The focal length and diameter of the objective lens are respectively greater than the focal length and diameter of the eyepiece.
- The objective lens is fitted at one end of a long metal tube. A metal tube of smaller diameter is fitted in this metal tube and the eyepiece is fitted at the outer end of the smaller tube.
- With the help of a screw it is possible to change the distance between the eyepiece and the objective lens by sliding the tube fitted with the eyepiece.
- The principal axes of the objective lens and the eyepiece are along the same line. A telescope is usually mounted on a stand.

Working:

- When the objective lens is pointed towards the distant object to be observed, the rays of light from the distant object, which are almost parallel to each other, pass through the objective lens.
- The objective lens collects maximum amount of light as it is large in size. It forms a real, inverted and diminished image in the focal plane of the objective lens.
- Now, the position of the eyepiece is adjusted such that this image falls just within the focal length of the eyepiece and serves as the object for the eyepiece which works as a simple microscope.
- The final image is highly magnified, virtual, on the same side as that of the object and inverted with respect to the original object.
- The final image can be observed by keeping the eye close to the eyepiece. If the image formed by the objective lens lies in the focal plane of the eyepiece, the final image is formed at infinity.



Refracting Telescope



Use of a telescope

A telescope is used to observe a distant object such as mountain, moon, planet, star in the magnified form.



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2024 - 25

Board: Maharashtra

METALLURGY

Physical Properties of Metal:

- (1) Solid State
- (2) Lustre
- (3) Ductility
- (4) Malleability
- (5) Good conductors of heat
- (6) Good conductors of electricity
- (7) Hard
- (8) Sonority
- (9) High Melting Point and Boiling Point

Physical Properties of Nonmetals:

- (1) Gas or solid state
- (2) Dull
- (3) Bad conductor of heat
- (4) Bad conductor of electricity
- (5) Soft
- (6) Non sonorous
- (7) Low melting point and low boiling point

Chemical Properties of Metals:

- (1) Reaction of metals with oxygen
- (2) Reaction of metals with water
- (3) Reaction of metals with oxygen
- (4) Reaction of metals with nitric acid
- (5) Reaction of metals with salts of other metals
- (6) Reaction of metals with nonmetals

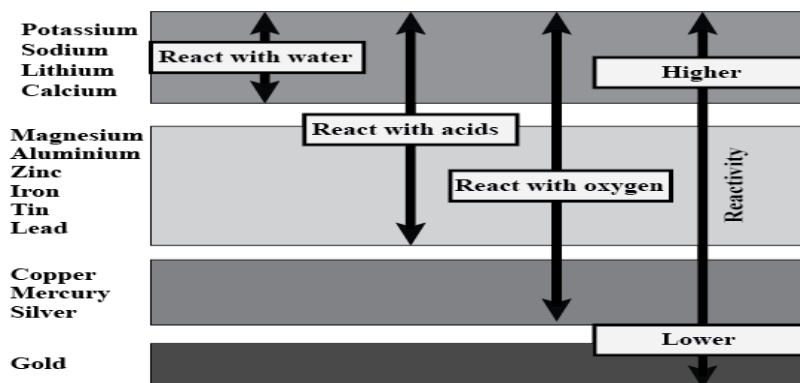
Chemical Properties of Nonmetals:

- (1) Reaction of nonmetals with oxygen
- (2) Reaction of nonmetals with water
- (3) Reaction of dilute acids with nonmetals
- (4) Reaction of nonmetals with hydrogen

Aqua Regia: Aqua regia is a highly corrosive and fuming liquid. It is one of the few reagents which can dissolve the noble metals like gold and platinum. Aqua regia is freshly prepared by mixing concentrated hydrochloric acid and concentrated nitric acid in the ratio 3 : 1.



Reactivity of Metals:



Reactivity series of metals.

Noble gases (Like helium, neon, argon) do not take part in the chemical reactions.

Nonmetals are also called **electronegative elements**, as they form negatively charged ions by accepting electron.

The compounds formed from two units, namely cation and anion are called **ionic compounds**.

The cation and anion being oppositely charged, there is an electrostatic force of attraction between them.

This force of attraction between cation and anion is called as the **ionic bond**.

General Properties of Ionic Compounds:

1. Exist in solid state and are hard.
2. Ionic compounds are brittle.
3. Melting and boiling points are high.
4. Ionic compounds are water soluble.
5. The ionic compounds cannot conduct electricity when in solid state. In the fused/molten state they can conduct electricity.

The science and technology regarding to extraction of metals from ores and their purification for the use is called **metallurgy**.

The compounds of metals that occur in nature along with the impurities are called **minerals**.

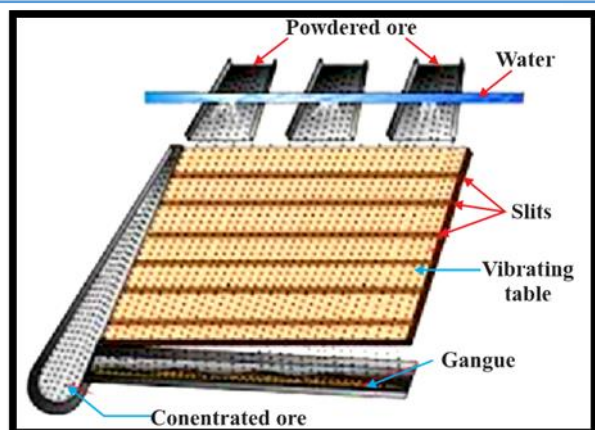
CONCENTRATION OF ORES

The process of separating gangue from the ores is called **concentration of ores**.

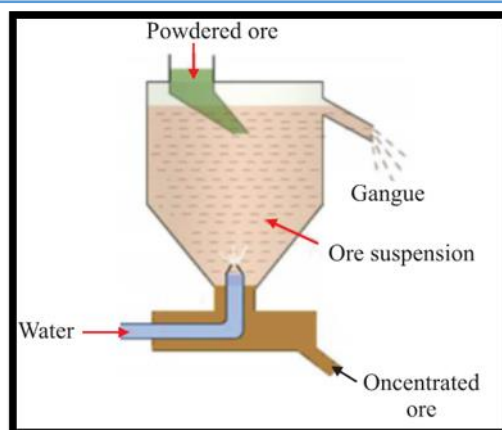
Concentration of ore basically means cleaning of ore's impurities such as silica , rocky particle, mud etc.

Different Method for Concentration of Ore:

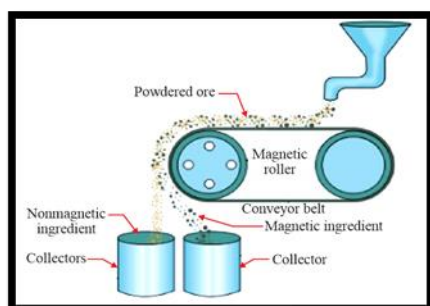
- (a) Separation based on gravitation
 - (1) Wilfley table method
 - (2) Hydraulic separation method
- (b) Magnetic separation Method
- (c) Froth floatation method
- (d) Leaching



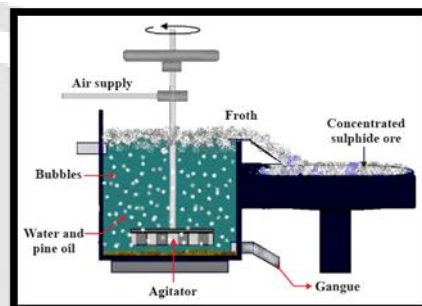
Wilfley table method



Hydraulic Separation



Magnetic Separation



Froth Floatation method

Leaching:

In this method the ore is soaked in a certain solution for a long time. The ore dissolves in that solution due to a specific chemical reaction. The gangue, however, does not react and therefore does not dissolve in that solution. So it can be separated.

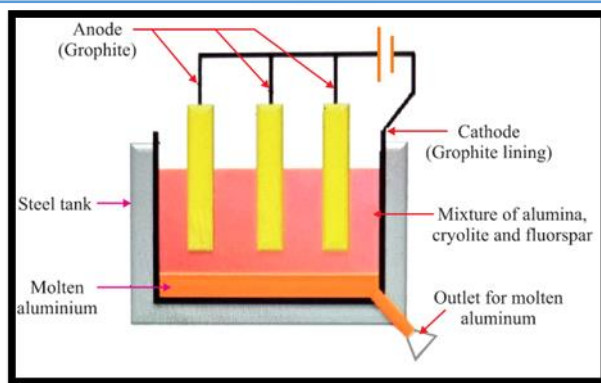
Water does not stick to Colocasia leaves. Similarly, water does not stick to wax. On the other hand common salt or soap stick to water that is, they get wetted by water.

2. Extraction of metals:

Extraction of Aluminium.

The metals at the top of the reactivity are highly reactive. For example, potassium, sodium, aluminium are reactive metals.

Reactive metals have large capacity to form cations by losing the electrons in their outermost shell. Their extraction has to be done by electrolytic reduction.

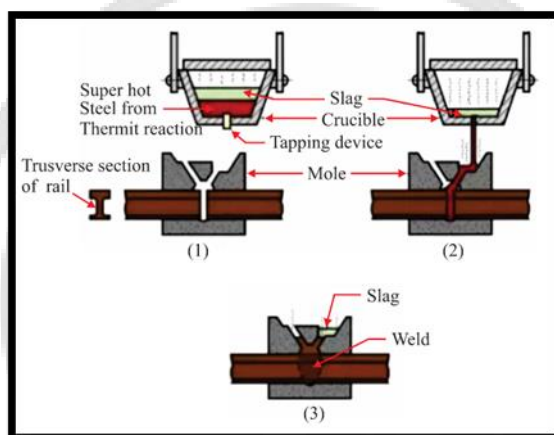


Extraction of aluminium

3. Extraction of moderately reactive metals:

The process of strongly heating the sulphide ores in air to transform them into oxides is called **roasting**.

The process of strongly heating the carbonate ores in a limited supply of air to transform them into oxides is called **calcination**.



4. Extraction of less reactive metals:

The metals at the bottom of the reactivity series of metals are less reactive. That is why they are found in free state in nature. For example gold, silver, platinum. The reserves of copper in free state are very few.

Refining of metals:

The process of obtaining pure metal from impure metal is called refining.

Corrosion of metals

- (1) Iron-Rusting
- (2) Copper- patination
- (3) Silver-Tarnish
- (4) Aluminium

Prevention of corrosion:

Galvanizing:

In this method a thin layer of zinc is applied to prevent corrosion of iron or steel.

Tinning:

In this method a layer of molten tin is deposited on metals.

Anodization:



In this method metals like copper, aluminium are coated with a thin and strong layer of their oxides by means of electrolysis.

Electroplating:

In this method a less reactive metal is coated on a more reactive metal by electrolysis.

Alloying:

The main intention behind this is to decrease the intensity of corrosion of metals.

The homogenous mixture formed by mixing a metal with other metals or nonmetals in certain proportion is called an **alloy**.

When one of the metals in an alloy is mercury the alloy is called **amalgam**. Silver amalgam was earlier used by dentists.



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2024-25

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CARBON AND COMPOUNDS

1. CARBON AND COMPOUNDS

- Intermolecular attractive forces are weak in carbon compounds.
- Generally, most of the carbon compounds are found to be bad conductors of electricity.
- Carbon has a unique ability to form strong covalent bonds with other carbon atoms. This property of carbon is called **catenation** power.
- The compounds which contain carbon and hydrogen as the only two elements are called **hydrocarbons**.
- The carbon compounds having a single bond between two carbon atoms are called **saturated compounds**.
- Saturated hydrocarbons are also called '**Alkanes**'.
- The carbon compounds having a double bond or triple bond between two carbon atoms are called **unsaturated compounds**.
- The unsaturated hydrocarbons containing a carbon-carbon double bond are called '**Alkenes**'.
- The unsaturated hydrocarbons whose structures contain a carbon-carbon triple bond are called '**Alkynes**'.
- In the course of millions of years the reserves of crude oil were formed from the dead organisms buried under the sea floor.
- This crude oil and natural gas are now recovered from the oil wells. The natural gas is mainly methane. The crude oil is a complex mixture of thousands of different compounds. It mainly contains various hydrocarbons.
- Various useful components such as CNG, LPG, petrol (gasoline), rockel, diesel, engine oil, lubricant, etc. are obtained by separation of crude oil using fractional distillation.

Hetero atom	Functional group		
	Name	Structural formula	Condensed structural formula
Halogen (chlorine, bromine, iodine)	Halo (chromo/bromo/iodo)	-X (-Cl, -Br, -I)	-X (-Cl, -Br, -I)
Oxygen	1. Alcohol	-O-H	-OH
	2. Aldehyde	$\begin{array}{c} \text{O} \\ \\ -\text{C} - \text{H} \end{array}$	-CHO
	3. Ketone	$\begin{array}{c} \text{O} \\ \\ -\text{C} - \end{array}$	-CO-
	4. Carboxylic acid	$\begin{array}{c} \text{O} \\ \\ -\text{C} - \text{O} - \text{H} \end{array}$	-COOH
	5. Ether	-O-	-O-
	6. Ester	$\begin{array}{c} \text{O} \\ \\ -\text{C} - \text{O} - \end{array}$	-COO-
Nitrogen	Amine	$\begin{array}{c} -\text{N} - \text{H} \\ \\ \text{H} \end{array}$	-NH ₂



The series of compounds formed by joining the same functional group in the place of a particular hydrogen atom on the chains having sequentially increasing length is called **homologous series**.

General molecular formula for the homologous series of

1. Alkanes - C_nH_{2n+2}
2. Alkenes - C_nH_{2n}
3. Alkynes - C_nH_{2n-2}

There are three units in the IUPAC name of any carbon compound:

prefix - parent – suffix

- The process of burning carbon and its compounds in excess of oxygen is known as **combustion**.
- The process of burning carbon and its compounds in excess of oxygen is known as **combustion**.
- Hydrocarbons as well as most of the carbon compounds undergo combustion in presence of oxygen to emit heat and light and form carbon dioxide and water as the common products.
- **Ethanol** being good solvent, it is used in medicines such as tincture iodine (solution of iodine and ethanol), cough mixture and also in many tonics.
- Methanol (CH_3OH), the lower homologue of ethanol, is poisonous, and intake of its small quantity can affect vision and at times can be lethal.
- To prevent the misuse of the important commercial solvent ethanol, it is mixed with the poisonous methanol. Such ethanol is called **denatured spirit**. A blue dye is also added to it, so that it is easily recognised.
- Ethanol is a clean fuel. Therefore, in some countries it is used as an additive to increase the efficiency of petrol. Such a fuel is called **gasohol**.
- Ethanoic acid is named as 'glacial acetic acid'.
- The giant carbon molecules formed by hundreds of thousands of atoms are called **macromolecules**.
- A macromolecule formed by regular repetition of a small unit is called **polymer**.
- The small unit that repeats regularly to form a polymer is called **monomer**.
- The reaction by which monomer molecules are converted into a polymer is called **polymerization**.

Polymer	Name of the Monomer	Occurrence
Polysaccharide	Glucose	Starch
Cellulose	Glucose	Wood (cell walls of plant cells)
Proteins	Alpha amino acids	Muscles, hair, enzymes, skin, egg
D.N.A.	Nucleotide (base-coxyribosc-phosphatc)	Chromosomes of animals
R.N.A.	Nucleotide (base-ribosc-phosphatc)	Chromosomes of plants
Rubber	Isoprene $\begin{array}{c} CH_2 = C - CH = CH_2 \\ \\ CH_3 \end{array}$	Latex of rubber tree





2024-25

SPACE MISSION

Sky

The visible portion of the atmosphere and outer space seen by simple eyes, without any equipment from the earth, is known as the sky.

Space

The infinite three-dimensional expanse in which the Solar system, stars, celestial bodies, galaxies and the endless Universe exist is known as space.

- Both sky and space lack a definite boundary.
- The sky is a very tiny part of space.

Objects in the Solar system

- Our Solar system is a very tiny part of a huge Galaxy-Milky Way.
- The Sun is at the centre of the Solar system. Sun is a star.
- Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune are planets in our Solar system. These planets revolve around the Sun.
- Some of these planets have their own natural satellites.
- Besides, there are asteroids, meteoroids, comets, and meteors in the Solar system.

Satellite

An astronomical object orbiting any planet of our Solar system is called a satellite.

- Mercury and Venus have no satellites.
- Some planets have more than one satellite. E.g. Jupiter has 69 satellites.
- The earth has one natural satellite called the moon.

Telescopes orbiting around the earth

The following three types of telescopes are orbiting around the earth :

- (i) Optical Refracting Telescope.
- (ii) Optical Reflecting Telescope.
- (iii) Radio Telescope.

Necessary of Telescopes to put in the space

Visible light and radio waves emitted by celestial bodies in space pass through the atmosphere before reaching the earth's surface.

- During this journey, some light is absorbed by the atmosphere. Hence, the intensity of the light reaching the earth's surface decreases.
- Besides, temperature and air pressure cause the atmospheric turbulence. Hence, light rays change their path, resulting in a change in the position of the image of a celestial body.
- City lights during night, and bright sunlight during day also put limitations on usage of optical telescopes on the earth.
- To minimize these problems, optical telescopes are situated on mountain top, away from inhabited places. However, limitations caused by the atmosphere still persist.

To get rid of these problems scientists have successfully launched telescopes in space. Images obtained by these telescopes are brighter and clearer than those obtained by the telescopes located on the earth's surface.

Space mission

There are 4 specific objectives of a space mission :

- To launch artificial satellites in the earth's orbit for study and research.
- To launch artificial satellites in the earth's orbit for various purposes like telecommunication, weather study, radio and TV programme transmission, etc.
- To send artificial satellites beyond the earth's orbit to observe, study and collect the information about other planets, meteors, meteoroids, asteroids and comets.
- To understand the part of the Universe beyond the Solar system.



Importance of space mission

Space missions are now essential to understand the origin and evolution of our solar system as well as to study the Universe beyond the Solar system. Space missions have given us many benefits and made our life simpler.

It is because of space missions that we are getting,

- Real-time information, communication across the globe.
- Entertainment and receiving the abundant information from the desk at any place.
- Access to huge information available on the internet.
- Management of natural resources.
- Advanced alerts and Management of some natural calamities and Pollution.
- Satellite surveillance for defence purpose.
- Access to various activities like trade, tourism and navigation.

Artificial satellites

A manmade object orbiting the earth or any other planet is called an artificial satellite.

- Satellites work on solar energy and hence photovoltaic panels are attached on both sides of the satellite, which look like wings.
- Satellites are installed with various transmitters and other equipment to receive and transmit signals between the earth and the satellites.

Classification of artificial satellites

(1) Weather satellite

To study and predict the weather conditions.

- It records temperature, air pressure, wind direction, humidity, cloud cover, etc. this information is sent to the space research station on the earth, and then with this information weather forecast is made.
- The names of Indian satellite: INSAT and GSAT. Launcher: GSLV.

(2) Communication satellite

To communicate between different locations in the world.

- Many artificial satellites placed at various locations in the earth's orbit are well interconnected and help us to have communication with any place, from anywhere, at anytime and in any form including voicemail, email, photographs, audio mail, etc.
 - In order to establish communication between different places on the earth through mobile phones or computer assisted internet, communication satellites are used.
- The names of Indian satellite: INSAT and GSAT. Launcher: GSLV.

(3) Broadcasting satellite

To telecast television and radio programmes.

- Due to this satellite we can have access to information about current incidents, events, programmes, sports and other events right from drawing room.
- The names of Indian satellite: INSAT and GSAT. Launcher: GSLV.

(4) Navigational satellite

To locate the place precisely in terms of its latitude and longitude and assist the surface, water and air transportation.

- These satellites also assist the user with current live maps as well as real time traffic conditions.
- The names of Indian satellite: IRNSS. Launcher : PSLV.

(5) Military satellite

Collecting information regarding security aspects including international borders.

- Satellites help to monitor all movements of neighbouring countries or enemy countries.
- Military satellites also help to guide the missiles effectively.



(6) Earth observation satellite

To study various aspects of the earth's surface including forests, oceans, polar regions and collect information about natural calamities.

- These satellites also help us to collect the information about the resources, their management, continuous observation about a natural phenomenon and the changes within it.
- The names of Indian satellite: IRS. Launcher : PSLV.

(7) Other satellites

Apart from these various satellites, certain satellites for specific works or purposes are also sent in the space.

Examples:

- India has sent EDUSAT for educational purpose.
- CARTOSAT for surveys and map making.
- Satellites with telescopes, like Hubble telescope or a satellite like International Space Station help to explore the universe.
- ISS (International Space Station) provides a temporary residence where astronauts can stay for a certain short or long period and can undertake the research and study space activities

Orbit of the artificial satellite

Orbit of a satellite is its path around the earth.

- The function of a satellite decides its orbit.
- The orbit can be circular or elliptical. The orbital plane of a satellite can be the equatorial plane of the earth or a plane at an angle to the equatorial plane.

Classification of satellites based on height

Based on the height of the satellite from the earth's surface, satellites are classified as follows :

- High Earth Orbit Satellites
- Medium Earth Orbit Satellites
- Low Earth Orbit Satellites

High Earth Orbit Satellite (HEO)

A satellite orbiting at a height equal to or greater than 35780km above the earth's surface is called High Earth Orbit satellite.

- If such a satellite revolves in the equatorial plane of the earth, in the same sense as the earth's rotation and has period equal to that of the earth's rotation, it will appear stationary with respect to a place on the earth. This satellite is, therefore, called geostationary or geosynchronous satellite.
- Such satellites are used for climatology, telecommunication, transmission of TV and radio programmes, etc.

Medium Earth Orbit Satellite (MEO)

A satellite orbiting at a height between 2000 km and 35780 km above the earth's surface is called a Medium Earth Orbit satellite.

- The orbital path of such a satellite is normally elliptical and passes through the North and the South polar regions. These satellites are useful for navigation.

Low Earth Orbit Satellite (LEO)

A satellite orbiting at a height between 180 km and 2000 km above the earth's surface is called a Low Earth Orbit satellite.

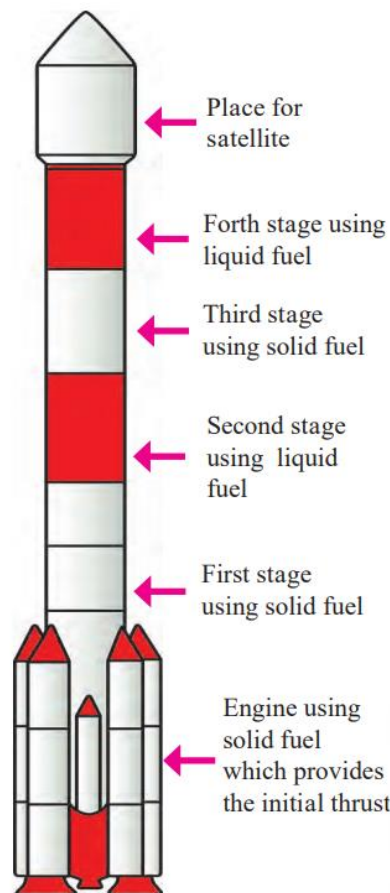
- Weather satellites, and International Space Station are low Earth orbit satellites.

Satellite Launch Vehicle

- A Satellite Launch Vehicle carries an artificial satellite to a desirable height above the earth's surface and places it in the desired orbit. It needs a specific velocity as well as a thrust to reach to the desired height.
- The velocity and thrust of a Satellite Launch Vehicle depend on the weight and orbital height of the satellite. Accordingly, the structure of the launch vehicle is decided and designed.



- The weight of the fuel also contributes a major portion in the total weight of the launch vehicle. This also influences the structure of the launch vehicle.
- In order to use the fuel optimally, multiple stage launch vehicles are now designed and used.
- As the journey of the launch vehicle progresses, the vehicle achieves a specific velocity and a certain height. The fuel in the first stage is exhausted and the empty fuel tank gets detached from the main body of the launch vehicle and falls back into a sea or on unpopulated land.
- As the fuel in the first stage is exhausted, the fuel in the second stage is ignited.
- However, the weight of the launch vehicle is now less than what it was earlier and hence it can move with higher velocity.
- Most of the launch vehicles are made up of two or more stages.
- Polar Satellite Launch Vehicle, that is, PSLV, developed by ISRO (Indian Space Research Organisation) has four stages.
- Launch vehicles are very costly as they can be used only once.
- To overcome this problem, USA has developed space shuttles which can be reused many times.
- Space shuttles can be used for both, (a) launching a satellite in space and (b) journey of astronauts.



Structure of PSLV made by ISR

Space missions away from the earth

- (1) Centripetal force on a satellite = Gravitational force exerted by the earth on the satellite on

$$\therefore \frac{mg_c^2}{(R+h)} = \frac{GMm}{(R+h)^2}$$

m : mass of the satellite, M : mass of the earth,

R : radius of the earth, h : height of the satellite from the surface of the earth,

v_c : critical velocity of the satellite, G : gravitational constant

$$v_c = \sqrt{\frac{GM}{R+h}}$$



Thus, if the value of h changes, the value of v_c also changes. It means a satellite needs to be given a specific velocity (in the tangential direction) to keep it revolving in a specific orbit.

Hence, if the height of the satellite from the surface of the earth increases, its velocity decreases.

- (2) The orbital duration of the satellite, i.e., the time required for one revolution of the satellite around the earth

$$T = 2\pi r / v_c$$

where $r = (R + h)$ and v_c = critical velocity

- (3) The escape velocity of a body is the minimum velocity with which it should be projected from the earth's surface, so that it can escape the influence of the earth's gravitational field.

$$\text{Escape velocity } v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

where v_{esc} = Escape velocity, G : gravitational constant, M : mass of the planet,

R = radius of the planet

The escape velocity on the moon is less than that on the earth.

Moon mission

- (i) As of now, only Russia, USA, European Union, China, Japan and India have successfully undertaken moon missions.
- (ii) Russia executed 15 moon missions between 1959 and 1976. Of these, last four missions brought the stone samples from the moon for study and analysis. However all these missions were unmanned.
- (iii) USA executed moon missions between 1962 and 1972. Some of these missions were unmanned. However, the historic moon mission took place on 20th July 1969, when American astronaut Neil Armstrong became the first human to step on the moon.
- (iv) Indian Space Research Organisation (ISRO) successfully launched Chandrayaan-I and placed it in the orbit of the moon. It sent useful information to the earth for about a year. The most important discovery made during the mission was the presence of water on the moon's surface. India was the first country to discover this.

Mars mission

- (i) Many nations have sent space crafts towards Mars, but only few of these missions have been successful.
- (ii) However, performance of Mangalyaan, the Indian spacecraft sent by ISRO towards Mars was remarkable. Mangalyaan was launched in November 2013 and was placed in the orbit of Mars successfully in September 2014. It has obtained useful information about the surface and atmosphere of Mars.

Other missions

- Some space crafts orbited the planets, some have landed on some planets and some just observed the planets, passed near them and went further to study other celestial bodies.
- Some space crafts were sent specifically to study asteroids and comets. Some space crafts have successfully brought dust and stone samples from the asteroids for study.
- All these space missions are very useful in getting information and helping us in clarifying our concepts about the origin of the earth and the Solar system.

Contribution of ISRO

- (1) India has indigenously built various launchers which can put the satellites having the mass upto 2500 kg in the orbit.
- (2) Indian Space Research Organisation (ISRO) has designed and built two important launchers; Polar Satellite Launch Vehicle (PSLV) and Geo Satellite Launch Vehicle (GSLV).
- (3) Successful space missions as well as scientific and technological accomplishments by India in space technology have made a significant contribution in the national and social development of our country.
- (4) Many satellites in INSAT and GSAT series are active in telecommunication, television broadcasting, meteorological services, disaster management and monitoring and management of natural resources.
- (5) EDUSAT is used specifically for education while satellites in IRNSS series are used for navigation.



- (6) Thumba, Sriharikota and Chandipur are Indian satellite launch centres.
- (7) Vikram Sarabhai Space Centre at Thiruvananthapuram, Satish Dhawan Space Research Centre at Sriharikota and Space Application Centre at Ahmadabad are space research organisations of India.

Management of space debris

Though very few countries in the world have the capability to launch satellites in orbit, the international cooperation and commercial prospects have led to many other countries having their own satellites in space.

- Due to failed launching, there are many satellites and parts of launchers, which are roaming freely in different orbits around the earth.
- In a way, they form space debris (fragments of satellites and launchers) orbiting in uncontrolled manner.
- By an estimate there are about two crore small and large pieces of satellites and launchers revolving around the earth.
- These debris certainly pose a threat to the current functional satellites, space shuttles and space stations.
- If this collection of fragments remains unchecked and uncontrolled, it is likely to be risky to all future launching of satellites and space shuttles and there is a growing possibility of space accidents.
- Research and dedicated efforts are on to find the means and ways of managing the space debris.



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